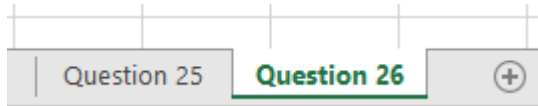


CMPG222 – Practical assignment 5

Dear students

Please do the following assignments at the end of **Chapter 6: 25 and 26**. The LOs (learning objectives) addressed by the assignments are indicated as part of the assignment. To complete the tasks, you must study Section 6.3 in the textbook. Please note that the data files you must use to do the assignments are shown to the left of the work in the book. These files are available on eFundi under the Resources tool. Copy the relevant data to your sheet and use the data to answer the assignment. Submit the answers to these assignments as **a single Microsoft Excel document** on the eFundi assignment. Create a separate sheet for each of the four questions as follows:



Also, remember to show the sub-sections of each question clearly in each sheet, e.g.:

Question 25(a)				
The answer to Question 25(a) should appear here.				
Question 25(b)				
The answer to Question 25(b) should appear here.				
Question 25(c)				
The answer to Question 25(c) should appear here.				
Question 25(d)				
The answer to Question 25(d) should appear here.				
Question 25(e)				
The answer to Question 25(e) should appear here.				

Kind regards

The CMPG222 lecturers

- 25. Association Rules for Bookstore Transactions.** Leggere, an Internet book retailer, is interested in better understanding the purchase decisions of its customers. For a set of 1,604 customer transactions, it has categorized the individual book purchases comprising those transactions into one or more of the following categories: Novels, Willa Bean series, Cooking Books, Bob Villa Do-It-Yourself, Youth Fantasy, Art Books, Biography, Cooking Books by Mossimo Bottura, Harry Potter series, Florence Art Books, and Titian Art Books. Leggere has conducted association rules analysis on this data set and would like to analyze the output. Based on a minimum support of 10% and a minimum confidence of 50%, the table below shows the top 10 rules with respect to lift ratio. **LO 7**
- Explain why the fourth rule on the list is not helpful although it has 100% confidence and lift ratio of 1.861.
 - Explain how the confidence of 55.6% and lift ratio of 1.851 was computed for the rule “If a customer buys a novel, a cooking book, and a biography book, then they buy an art book.” Interpret these quantities.
 - Based on these top 10 rules, what general insight can Leggere gain on the purchase habits of these customers?
 - What will be the effect on the rules generated if Leggere decreases the minimum support and reruns the association rules analysis?
 - What will be the effect on the rules generated if Leggere decreases the minimum confidence and reruns the association rules analysis?

Antecedent	Consequent	Support of A	Support of C	Support of A & C	Confidence	Lift Ratio
Novels, Cooking, Art	Biography	0.1552	0.3454	0.1016	0.655	1.895
Novels, Cooking, BobVilla	Art	0.1827	0.3005	0.1022	0.560	1.863
Novels, BotturaCooking	Cooking	0.1060	0.5374	0.1060	1.000	1.861
BotturaCooking	Cooking	0.1415	0.5374	0.1415	1.000	1.861
Novels, Cooking, Biography	Art	0.1827	0.3005	0.1016	0.556	1.851
Novels, Cooking, Art	BobVilla	0.1552	0.3635	0.1022	0.659	1.812
Cooking, BobVilla	Art	0.2363	0.3005	0.1278	0.541	1.800
Novels, Art	Biography	0.2007	0.3454	0.1241	0.618	1.789
Novels, YouthFantasy	Biography	0.1895	0.3454	0.1166	0.615	1.781
Cooking, Art	Biography	0.2082	0.3454	0.1272	0.611	1.768

- 26. Congressional Voting Records.** Freelance reporter Irwin Fletcher is examining the historical voting records of members of the U.S. Congress. For 175 representatives, Irwin has collected the voting record (yes or no) on 16 pieces of legislation. To examine the relationship between representatives' votes on different issues, Irwin has

conducted an association rules analysis with a minimum support of 40% and a minimum confidence of 90%.

The data included the following bills:

Budget: approve federal budget resolution

Contras: aid for Nicaraguan contra rebels

El_Salvador: aid to El Salvador

Missile: funding for M-X missile program

Physician: freeze physician fees

Religious: equal access to all religious groups at schools

Satellite: ban on anti-satellite weapons testing

The following table shows the top five rules with respect to lift ratio. The table displays representatives' decisions in a "bill-vote" format. For example, "Contras-y" indicates that the representative voted yes on a bill to support the Nicaraguan Contra rebels and "Physician-n" indicates a no vote on a bill to freeze physician fees. **LO 7**

Antecedent	Consequent	Support of A&C	Confidence	Lift Ratio
Contras-y, Physician-n, Satellite-y	El_Salvador-n	0.40	0.95	1.98
Contras-y, Missile-y	El_Salvador-n	0.40	0.91	1.91
Contras-y, Physician-n	El_Salvador-n	0.44	0.91	1.90
Missile-n, Religious-y	El_Salvador-y	0.40	0.93	1.90
Budget-y, Contras-y, Physician-n	El_Salvador-n	0.41	0.90	1.89

- Interpret the lift ratio of the first rule in the table.
- What is the probability that a representative votes no on El Salvador aid given that they vote yes to aid to Nicaraguan Contra rebels and yes to the M-X missile program?
- What is the probability that a representative votes no on El Salvador aid given that they vote no to the M-X missile program and yes to equal access to religious groups in schools?
- What is the probability that a randomly selected representative votes yes on El Salvador aid?